Lecture 3

Power Semiconductor Devices

A transistor is turned on by applying a gate control voltage. As long as the gate voltage is applied, the transistor remains turned on and it switches to off-state if the gate voltage is removed. The collector–emitter voltage of a bipolar transistor (simply BJT) depends on its base current. As a result, a significant amount of base current may be needed to drive a BJT switch into the low resistance saturation region. On the other hand, the drain–source voltage of a MOS-type (metal oxide semiconductor) transistor depends on its gate voltage and its gate current is negligible. As a result an MOSFET does not require any gate current, and the gate power to drive a MOSFET switch into the low resistance saturation region is negligible. A power semiconductor device with a MOS-type of gate control is preferable and the development of power devices technology is progressing accordingly.

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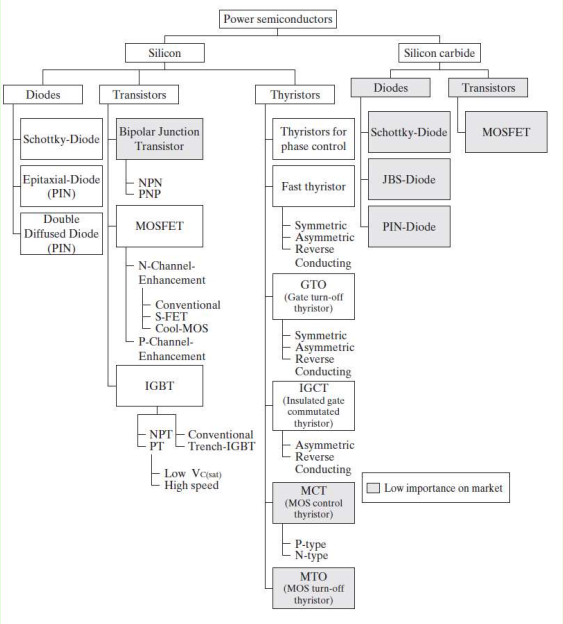
Power Semiconductor Devices

The ratings of power devices are continuously improving and one should check the available power devices. A superpower device should (1) have a zero on-state voltage, (2) withstand an infinite off-state voltage, (3) handle an infinite current, and (4) turn on and off in zero time, thereby having infinite switching speed.

With the development of SiC-based power devices, the switching time and the on-state resistance would be significantly reduced while the device voltage rating would increase almost 10 times. As a result, the applications of power devices are expected to change.

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Power Semiconductor Devices



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Power Semiconductor Devices

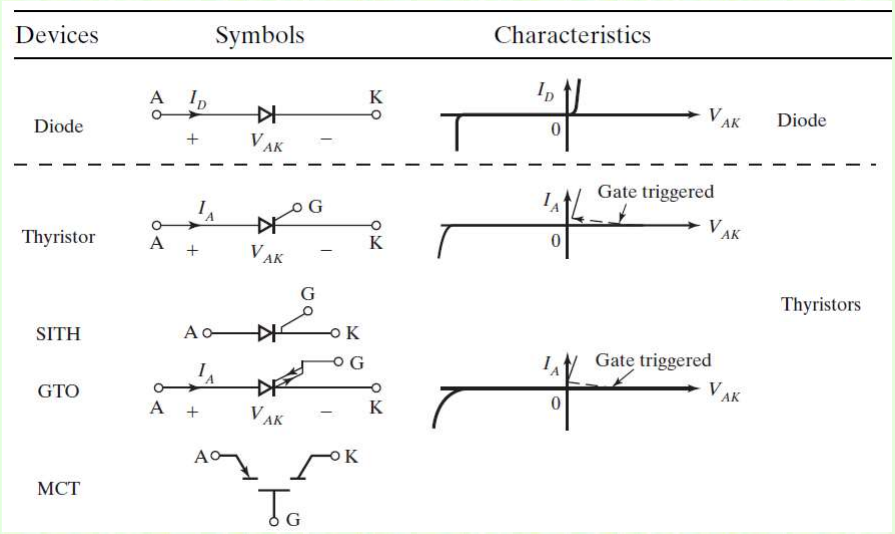
The power semiconductor switching devices can be classified on the basis of:

1. Uncontrolled turn-on and turn-off (e.g., diode);
2. Controlled turn-on and uncontrolled turn-off (e.g., SCR);
3. Controlled turn-on and -off characteristics (e.g., BJT, MOSFET, GTO, SITH, IGBT, SIT, MCT);
4. Continuous gate signal requirement (BJT, MOSFET, IGBT, SIT);
5. Pulse gate requirement (e.g., SCR, GTO, MCT);
6. Bipolar voltage-withstanding capability (SCR, GTO);
7. Unipolar voltage-withstanding capability (BJT, MOSFET, GTO, IGBT, MCT);
8. Bidirectional current capability (TRIAC, RCT);
9. Unidirectional current capability (SCR, GTO, BJT, MOSFET, MCT, IGBT, SITH, SIT, diode).

Static induction transistor/thyristor (SIT, SITh), Gate turn-off thyristor (GTO), MOS Turn-off (MTO)
Reverse conducting thyristors (RCT), MOS-controlled thyristor (MCT), Emitter turn-off thyristor (ETO)
Insulated-gate bipolar transistor (IGBT), Integrated gate-commutated thyristor (IGCT)

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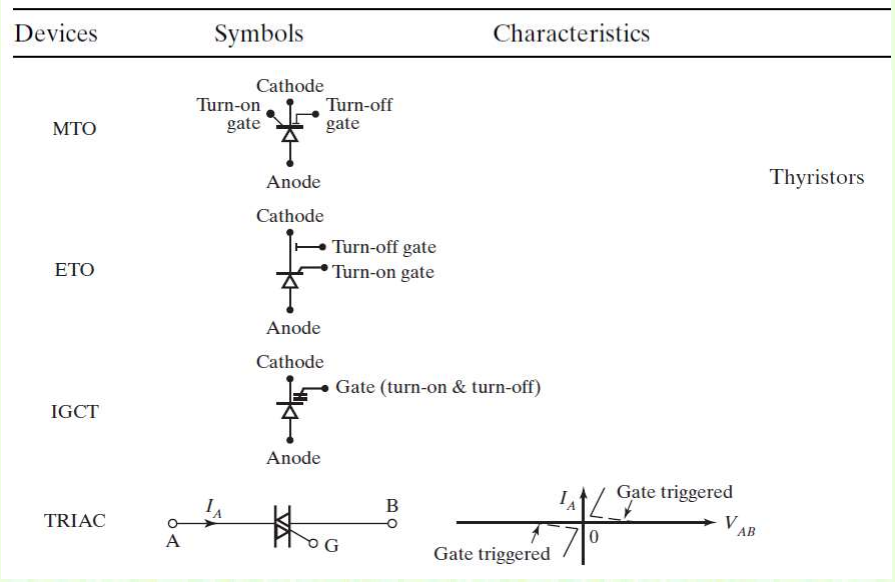
Power Semiconductor Devices



Static induction thyristor (SITh), Gate turn-off thyristor (GTO), MOS-controlled thyristor (MCT)

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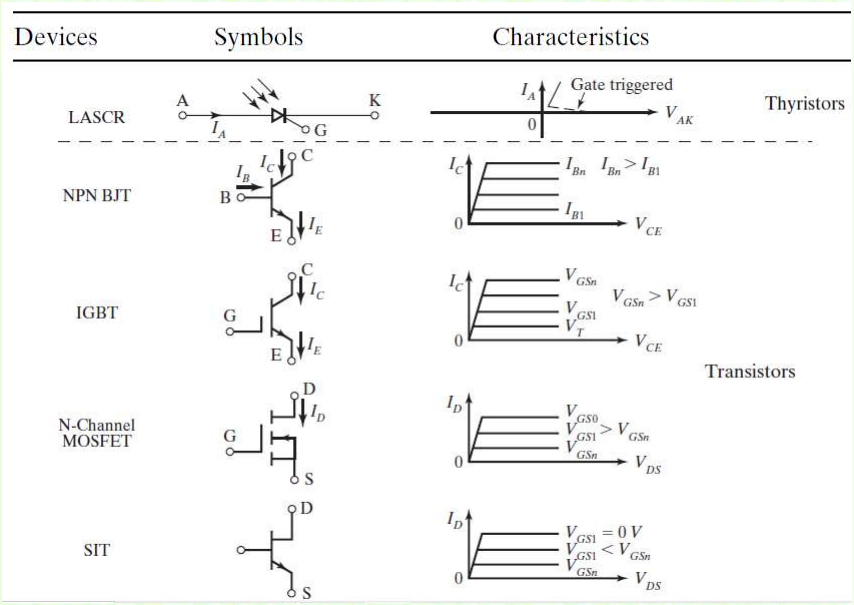
Power Semiconductor Devices



MOS Turn-off (MTO), Emitter turn-off thyristor (ETO), Integrated gate-commutated thyristor (IGCT)

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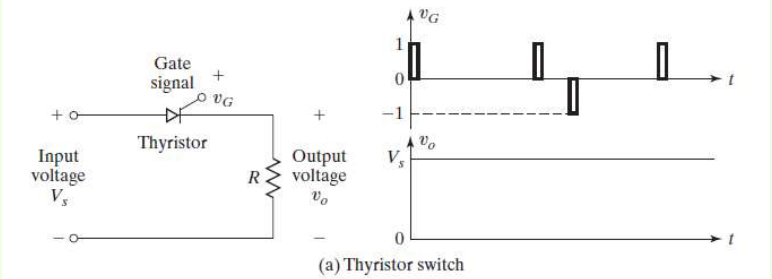
Power Semiconductor Devices



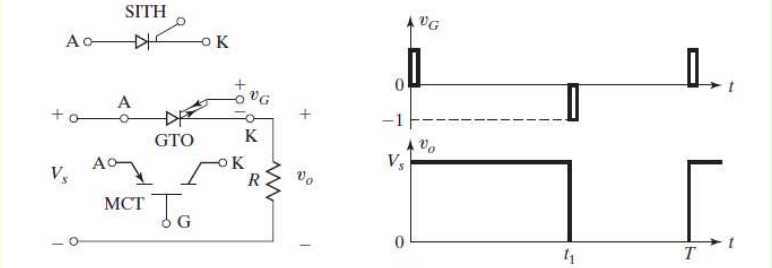
Light activated silicon controlled rectifier (LASCR), Static induction transistor (SIT)

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Control Characteristics of Power Devices



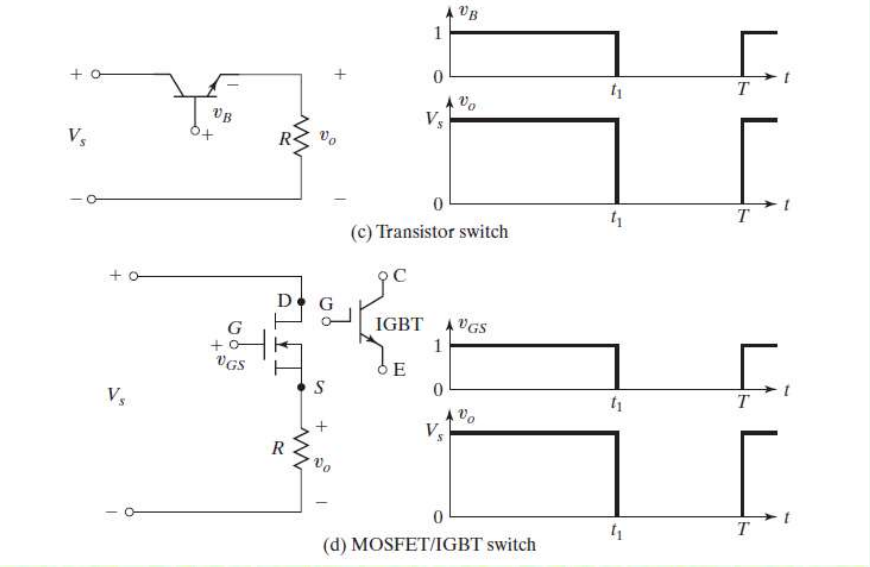
(a) Thyristor switch



(b) GTO/MTO/ETO/IGCT/MCT/SITH switch (For MCT, the polarity of V_G is reversed as shown)

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Control Characteristics of Power Devices



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Control Characteristics of Power Devices

TABLE 1.5 Switching Characteristics of Power Semiconductors

Device Type	Device	Continuous Gate	Pulse Gate	Controlled Turn-On	Controlled Turn-Off
Diodes	Power diode				
Transistors	BJT	x		x	x
	MOSFET	x		x	x
	COOLMOS	x		x	x
	IGBT	x		x	x
	SIT	x		x	x
Thyristors	SCR		x	x	
	RCT		x	x	
	TRIAC		x	x	
	GTO		x	x	x
	MTO		x	x	x
	ETO		x	x	x
	IGCT		x	x	x
	SITH		x	x	x
	MCT	x		x	x

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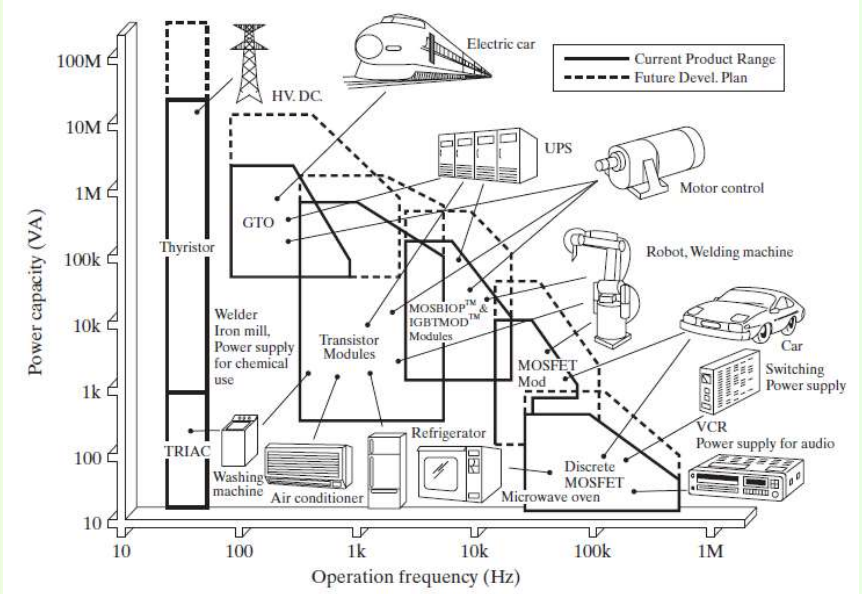
Control Characteristics of Power Devices

TABLE 1.5 Switching Characteristics of Power Semiconductors

Device Type	Device	Unipolar Voltage	Bipolar Voltage	Unidirectional Current	Bidirectional Current
Diodes	Power diode	x		x	
Transistors	BJT	x		x	
	MOSFET	x			x
	COOLMOS	x			x
	IGBT	x		x	
	SIT	x		x	
Thyristors	SCR		x	x	
	RCT		x		x
	TRIAC		x		x
	GTO		x	x	
	MTO		x	x	
	ETO		x	x	
	IGCT		x	x	
	SITH		x	x	
	MCT		x	x	

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Power Semiconductor Devices



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Device Choices

TABLE 1.6 Device Choices for Different Power Levels

Choices	Low Power	Medium Power	High Power
Power Range	Up to 2 kW	2 to 500 kW	More than 500 kW
Usual Converter Topologies	ac-dc, dc-dc	ac-dc, dc-dc, dc-ac	ac-dc, dc-ac
Typical Power Semiconductors	MOSFET	MOSFET, IGBT	IGBT, IGCT, thyristor
Technology Trend	High power density High efficiency	Small volume and weight Low cost and high efficiency	High nominal power of the converter High power quality and stability
Typical Applications	Low-power devices Appliances	Electric vehicles Photovoltaic roofing	Renewable energy Transportation Power distribution Industry

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Device Choices

The following guidelines in general can be used to select a device for most applications depending on the type of input supply.

For a dc input source:

1. Check if a power MOSFET can meet the voltage, the current, and the frequency of the intended applications.
2. If you cannot find a suitable power MOSFET, check if an IGBT can meet the voltage, the current, and the frequency of the intended applications.
3. If you cannot find a suitable power MOSFET or an IGBT, check if a GTO or an IGCT can meet the voltage, the current, and the frequency of the intended applications.

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Device Choices

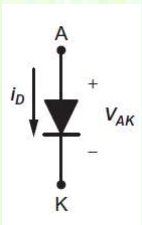
The following guidelines in general can be used to select a device for most applications depending on the type of input supply.

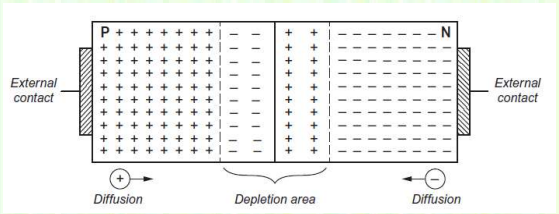
For AC input source:

1. Check if a TRIAC can meet the voltage, the current, and the frequency of the intended applications.
2. If you cannot find a suitable TRIAC, check if a thyristor can meet the voltage, the current, and the frequency of the intended applications.
3. If you cannot find a suitable TRIAC or a thyristor, you can use a diode rectifier to convert the ac source into a dc source. Check if an MOSFET or an IGBT can meet the voltage, the current, and the frequency of the intended applications.

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Power Electronic Devices
Power Diode



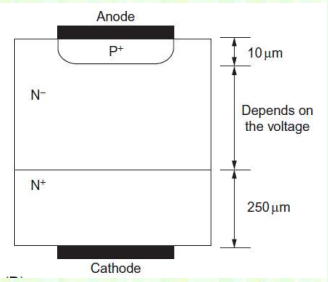


A power diode is a type of diode that is commonly used in power electronics circuits. Just like a regular diode, a power diode has two-terminals and conducts current in one direction. A power diode varies in construction from a standard diode to enable this higher current rating.

The ordinary signal diodes have a junction formed by p type semiconductor and n type semiconductor, the lead joining p-type is called the anode and the other side lead joining the n-type is called the cathode.

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Power Electronic Devices
Power Diode



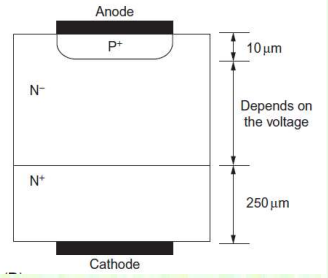
In power diodes, we have a junction formed between a heavily doped P+ and a lightly doped N– the layer which is epitaxially grown on a heavily doped N+ layer.

The N– layer is the key feature of the power diode which makes it suitable for high power applications. This layer is very lightly doped, almost intrinsic and hence the device is also known as a PIN diode, where i stands for intrinsic.

This is due to its light doping concentration, as we know that the thickness of the space charge region increases with a decrease in doping concentration.

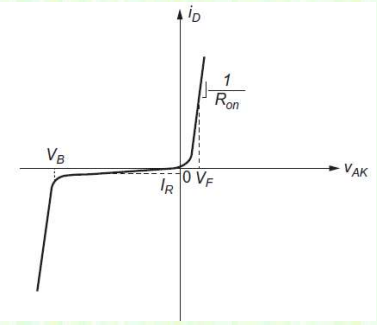
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Power Electronic Devices
Power Diode



This increased thickness of the depletion region or the space charge region helps the diode to block larger reverse-biased voltage and hence have a greater breakdown voltage.

However, adding this N– layer significantly increases the ohmic resistance of the diode leading to more heat generation during the forwarding conduction state. Hence power diodes come with various mountings for proper heat dissipation.



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Power Diode

Forward voltage, V_F: The voltage drop across the diode, from terminal A to K, at a specific current when the diode is forward-biased.

Breakdown voltage, V_B: The maximum reverse voltage across the diode before the diode starts reverse conduction and gets severe damaged. The point at Vb is also known as avalanche point.

Reverse current I_R: The current at a particular reverse voltage. This current is usually in the range of μA and hardly taken into account in analysis of power electronic circuits.

Conduction resistance R_{ON}: The equivalent resistance presented by the diode when forward-biased and conducting current. The inverse of this resistance is the slope of the curve in forward region. This characteristic is more relevant in high-voltage devices in which the semiconductor length is larger.

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